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| **Fiscal Year:** | FY 2015 | **Task Last Updated:** | FY 10/28/2015 |
| **PI Name:** | Stolc, Viktor Ph.D. | | |
| **Project Title:** | Electromagnetic Basis of Metabolism and Heredity | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | **TechPort:** | No | |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Microbiology | | |
| **Space Biology Cross-Element Discipline:** | (1) Reproductive Biology | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | viktor.stolc-1@nasa.gov | **Fax:** | FY |
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| **Organization Name:** | NASA Ames Research Center | | |
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| **City:** | Moffett Field | **State:** | CA |
| **Zip Code:** | 94035 | **Congressional District:** | 18 |
| **Comments:** | | | |
| **Project Type:** | Ground,New Investigation | **Solicitation / Funding Source:** | 2012 Space Biology NNH12ZTT001N |
| **Start Date:** | 08/01/2013 | **End Date:** | 05/31/2015 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 2 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 3 |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | 3 |
| **No. of Bachelor's Candidates:** | 4 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Smith, Jeffrey | **Contact Phone:** | 650-604-0880 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Extended to 5/31/2015 (had previously been extended to 12/31/2014) per A. Chu/ARC (Ed., 2/26/15)<br>NOTE: Extended to 12/31/2014 (original end date was 7/31/2014) per A. Chu/ARC (Ed., 5/7/14) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Freund, Friedemann ( SETI Institute ) | | |
| **Grant/Contract No.:** | Internal Project | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | We propose to investigate how the Earth's natural electromagnetic field in the ultralow to extremely low frequency (ULF/ELF) range interacts with crucial reaction steps in biochemical processes that have time constants in the ULF/ELF frequency range. We propose to address the temporal coordination of fundamental biochemical reactions in living cells, which are maintained by the sequential transfer of electrons in reduction-oxidation reactions. This redox cycle (also called the metabolic cycle) is known to control all cellular responses to environmental factors, including those caused by spaceflight-induced stress. In fact, the redox cycle is the fundamental biochemical process that controls the timing of all biochemical reactions in living cells, including energy production, RNA transcription, and DNA replication. Through this temporal coordination chemical conflict between the reductive and oxidative reactions is avoided. At the same time the coherent or interfering transfer of energy is enabled between the ambient natural electromagnetic (EM) field and living cells. We will study how the EM field controls the timing of the biochemical redox cycle and thereby controls heredity. Our main tool to carry out this work will be an enclosure (chamber) with an electronic circuit enabling in real-time, through a dynamic feedback loop, active attenuation, or cancelation or modulation of the ambient EM field. | | |
| **Rationale for HRP Directed Research:** | | | |

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| **Research Impact/Earth Benefits:** | The significance of the proposed work to confirm EM resonance as an environmental unit factor that affects, or perhaps even drives metabolism, growth, and genome integrity is that confirmation of this new understanding about the ambient Earth and spaceflight environments would enable the previously unattainable goal of mitigating the negative effects of spaceflight on human physiology beyond the lower Earth orbit (LEO). Therefore, successful execution of this proposal can enable human space exploration beyond LEO by advancing fundamental understanding of metabolism on Earth and in spaceflight. |
| **Task Progress:** | We have generated preliminary results during the last year that are consistent with the proposed hypothesis that the ambient electromagnetic (EM) field is an environmental unit factor for time-setting, a kind of “tuning fork” for universal biochemical reactions in living cells. Remarkably, we found that extremely small variation in Schumann Resonance (SR) modulates the electron flux through the electron transport chain in living organisms. Specifically, we have developed an EM attenuation chamber with a dynamic feedback loop electronics that enable selectable, real-time cancelation of the ambient EM field. Using this chamber, we have demonstrated that attenuation of the Earth’s ambient magnetic field not only reduced yeast cell division rate, as assayed by the budding index, but also significantly reduced basal lipid oxidation likely due to reduction of basal electron leakage from the electron transport chain (ETC). Over the course of this investigation, we have identified several systems, including stepper motors, rotary solenoid, homopolar motors/generators, that could have features similar to the ATPase. We have performed numerical simulation of the torque from SR to determine whether or not it could provide a fraction of the energy needed to overcome the potential barrier to ATP synthesis to exceed the activation energy. Our work demonstrated previously un-described evidence for quantum effects in biology by EM coupling below thermal noise at the universal electron transport chain (ETC) in vivo. We propose to further develop an electromagnetic chamber for arbitrary modulation of the ambient Schumann Resonance field, including its active cancellation, to enable reversible slowing of life processes without termination. The proposed suspended animation chamber is revolutionary because no other method can actively account for the quantum effects of SR on the fundamental life-sustaining process of electron transport in vivo. |
| **Bibliography Type:** | Description: (Last Updated: 07/01/2025) |
| **Articles in Peer-reviewed Journals** | Griko Y, Loftus D, Stolc V. "Metabolic suppression: A promising solution to unlock the future of space travel." Journal of Tourism & Hospitality. 2024 Jan 18;12(6):537. https://www.longdom.org/open-access/metabolic-suppression-a-promising-solution-to-unlock-the-future-of-space-travel-105320.html#ai , Jan-2024 |